Assignment 3: Advanced optics
Group and phase velocity, Lorentz model, Conductive materials

Marks: 25
Due date: *September 15, 2026*

1. (i) The phase velocity v of transverse waves in a crystal of atomic separation a is given by $v = \left(c\frac{\sin(ka/2)}{(ka/2)}\right)$, where k is the wave number and c is constant. Show that the value of the group velocity is $c\cos(ka/2)$. What is the limiting value of the group velocity for long wavelengths?
(ii) The dielectric constant of a gas at a wavelength λ is given by

$$\epsilon_r = \frac{c^2}{v^2} = A + \frac{B}{\lambda} - \frac{D}{\lambda^2}$$

,
where A, B and D are constants, c is the velocity of light in free space and v is its phase velocity. If the group velocity is V_g, show that $V_g\epsilon_r = v(A + B/2\lambda)$.
(iii) The relative permittivity of an ionized gas is given by $\epsilon_r = \frac{c^2}{v^2} = 1 - \left(\frac{\omega_e}{\omega}\right)^2$, where v is the phase velocity, c is the velocity of light and ω_e is the constant value of the electron plasma frequency. Show that this yields the dispersion relation $\omega^2 = \omega_e^2 + c^2k^2$, and that as $\omega \to \omega_e$, the phase velocity exceeds that of light, c, but that the group velocity (the velocity of energy transmission) is always less than c. [3+3+3]

2. Assuming a material for which $\sigma < \epsilon\omega$, find out the expressions for attenuation length, group velocity, magnetic field, and energy densities of electric and magnetic fields using the same approach we followed in class. Do you find any interesting observations or trends? [4+2]

3. Take an insulator whose resonance frequency is $\omega_0 = 6.4 \times 10^{14}$ rad/s, damping factor $= 0.28\omega_0$, plasma frequency $\omega_p = 7.8 \times 10^{16}$ rad/s.
(a) What is the electron density of this material?
For the following, use MATLAB or Python. Make sure to adjust the plots to include all the relevant features. Label the axes carefully, include appropriate units and indicate the position of ω_0. Calculate all local maxima and minima and include in your answer the frequency for which they occur as well as the value of the function at those points. Identify the Transmission-Absorption-Reflection-Transmission regions in all plots. Each item should have one plot (i.e., plot both quantities in the same graph). If necessary, assume that this material is surrounded by vacuum.
(b) Plot ϵ_r and ϵ_i, the real and imaginary parts of the complex permittivity.
(c) Plot n and κ, the real refractive index and the extinction coefficient. What is the phase velocity of light travelling in this material? (d) Plot R, the reflectivity. Also plot the transmittance T, but not in the absorption region (A).
(e) What happens - predominantly - to the individual frequency components of a polychromatic electromagnetic wave composed of soft X-rays ($\lambda_1 = 4.05$nm), red light ($\lambda_2 = 632.5$nm) and FM radio ($\lambda_3 = 1.4$m), when the wave hits this insulator?
[N.B.: Attach the plots in the order of the questions with proper label and legend. Write appropriate comments, if necessary] [1+9]